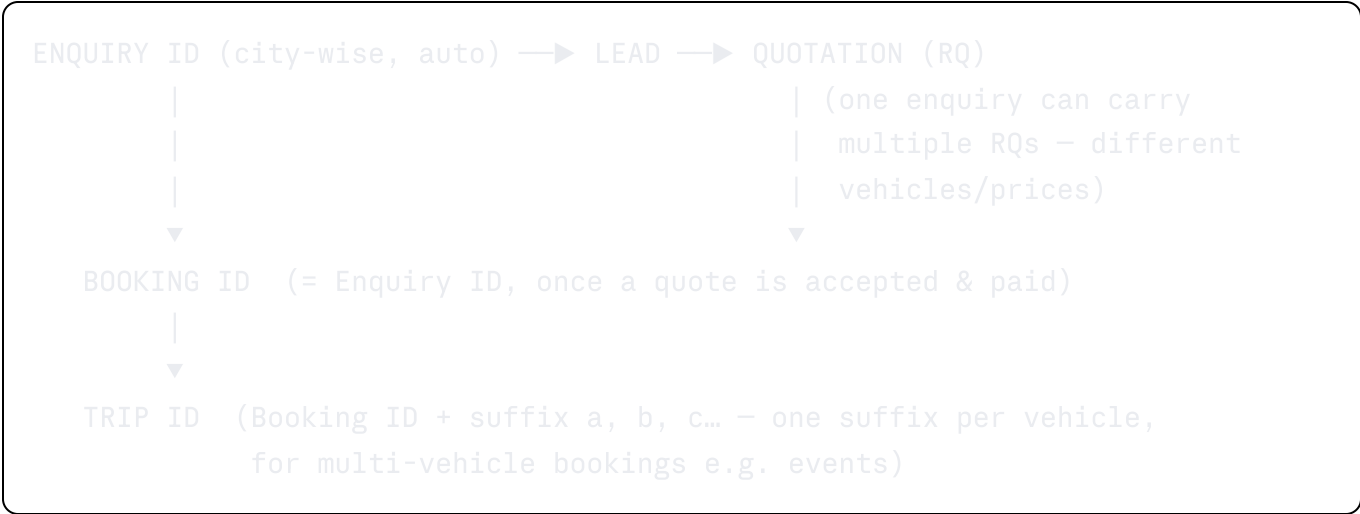


PINAAK Mobile App — Business Logic & RBAC Specification

Phase 1 scope: Customer · Vendor (BA) · Driver — UC internal roles excluded

1. Core Entity & ID Chain

Every transaction in PINAAK flows through one linked ID chain. This is the backbone every role-permission rule hangs off of.



Rule: A Customer, Vendor, or Driver never sees an ID that isn't linked to them. Visibility is always scoped by this chain, not by a flat "see everything of this type" permission.

2. Roles & Sub-Roles in This Phase

Role	Sub-roles / logins	Notes
Customer	Personal (1 login) · Corporate (3 logins: Booking Person, Admin, A/C) · Event (3 logins + Event Vehicle Scheduler)	Corporate/Event logins share one Customer ID but differ in what they can <i>do</i>
Vendor (BA)	Owner · Booking Manager · Ops Manager · A/C Manager	Up to 4 logins per vendor company, same Vendor ID
Driver	Single role	Attached to one Vendor; assigned to a specific Vehicle per Trip, not permanently

3. Business Logic Flow — Stage by Stage

Stage	What happens (system logic)	Customer	Vendor (BA)	Driver
Enquiry / Lead	Created on the UC side (call/WA/web/ads) — out of scope this phase	Receives quote notification once UC has qualified the lead	Not involved yet	Not involved
Quotation	UC creates one or more RQs against the enquiry (different vehicle/price options)	Views quote(s) in a cart-style screen, selects a variant, can request revision (budget remark) or confirm	Not involved	Not involved
Booking	Quote confirmed + payment (or advance) received → Booking ID generated	Gets booking confirmation, must acknowledge it before any driver/vehicle detail is released	Receives broadcast (if UC is sourcing vehicle) or direct assignment (if BA already tied to route); can Accept / Reject (with reason) / propose vehicle change	Not involved
Vehicle Assignment	BA accepts → BA assigns vehicle + driver	Sees "vehicle & driver being arranged" status only	Selects vehicle from own fleet, assigns driver, can request UC approval for a vehicle swap	Gets a new-trip notification once assigned
Pre-Trip	Duty Slip generated (BA + Driver), Vehicle Slip generated (to Customer)	Adds passenger list + pickup address (mandatory ~3 days ahead); sees driver & vehicle detail only after UC approval	Views/confirms duty slip; runs pre-dispatch checklist	Views duty slip (customer address/phone revealed only close to trip time — privacy rule, see §5)
Trip (live)	Trip ID active	Views live driver/vehicle	Views trip status (upcoming/ongoing)	Views customer's live

Stage	What happens (system logic)	Customer	Vendor (BA)	Driver
		location; receives OTP to hand to driver at pickup	for own fleet only	location; enters the OTP given by customer; captures start/end KM with photo
Payment	Booking amount %, advance, balance tracked automatically	Views total/paid/balance; pays via gateway; system blocks paying more than the booking amount	Views/confirms BA advance payment; raises GST bill after trip	Confirms customer payment collected (if COD-style component exists)
Feedback	Mandatory feedback with photo/video after trip closes	Submits feedback (vehicle + driver)	Views own feedback score/history	Not involved

4. RBAC Matrices

Legend: **V** = View · **C** = Create · **E** = Edit · **A** = Approve/Confirm · **—** = No access

4.1 Customer sub-roles

Module / Screen	Personal	Corp - Booking Person	Corp - Admin	Corp - A/C	Event - Booking Person	Event - Admin/Scheduler
View Quote	V	V, C(request revision)	V	V	V	V
Confirm Booking	A	A	V	—	A	A
Event Vehicle Scheduler (multi-vehicle single screen)	—	—	—	—	—	V, C, E
Passenger List / Pickup Address	C, E	C, E	V	—	C, E	C, E
Live Trip Tracking	V	V	V	—	V	V
Payment (view + pay)	V, C	V, C	V	V	V, C	V
Feedback	C	C	V	—	C	V
Multi-login visibility	n/a	Sees own actions + Admin's overrides	Sees all 3 logins' activity	Sees payment trail only	Sees own actions	Sees all bookings across the event

4.2 Vendor (BA) sub-roles

Module / Screen	Owner	Booking Manager	Ops Manager	A/C Manager
Vendor Profile / Enrolment	V, E	V	V	V
Vehicle Master (add/edit vehicle)	C, E, A	V	C, E	V
Broadcast — view & respond with rate	V, C	V, C	V	—
Booking Accept / Reject	A	A	V	—
Assign Driver + Vehicle to Trip	A	V	C, E, A	—
Pre-Dispatch Checklist	A	V	C, E, A	—
Payments (advance received, weekly status)	V	V	V	V, C
Raise GST Bill	A	—	—	C, A

4.3 Driver

Single role — no sub-roles — but two **data-visibility rules** apply (see §5):

Module / Screen	Access
My Trips (Upcoming / Completed / Cancelled)	V — own trips only
Duty Slip detail	V — customer contact fields time-gated
Customer Live Location	V — only once trip window opens
Enter OTP	C — one attempt flow, tied to Trip ID
Start/End KM + Photo capture	C — timestamp auto-captured, not editable
Driver Briefing checklist	V, A (acknowledge)

5. Business Rules That Must Be Enforced in the RBAC Layer (not just UI hiding)

1. **Time-gated visibility, not role-based hiding:** Driver only sees the customer's phone/address once the Duty Slip is issued and trip window is near — this is a *state* condition layered on top of role, not a permanent permission.

2. **Approval gates release information:** Customer never sees Driver/Vehicle detail until UC has approved the assignment — the Customer app must poll/subscribe to an "approved" flag, not just "assigned."
3. **Payment ceiling:** the app must hard-block (not just warn) a Customer from paying more than the current Booking Amount — this is a server-side rule the mobile client should mirror for UX but never rely on client-side alone.
4. **Multi-login = shared entity ID, split permissions:** Corporate/Event Customer logins and Vendor's 4 logins all resolve to the *same* Customer ID / Vendor ID in the data model. RBAC must be enforced per **login (user)**, **not per entity** — two people at the same company can have very different screens.
5. **Vendor can accept a booking without a vehicle number yet** (and add it later) — the Ops Manager permission to "Edit vehicle assignment post-acceptance" must remain open even after the Booking Manager has already approved.
6. **Driver is never permanently tied to a Vendor's specific vehicle** — assignment is per-Trip, so "my vehicle" on the Driver home screen must be trip-scoped, not a fixed profile field.

6. Screens Used in the Interactive Prototype

Role	Screens
Customer	Home (status) → Quote → Bookings → Trip (live) → Payments → Profile
Vendor	Home (broadcasts + new assignments) → Booking Detail (accept/reject/assign) → Vehicles → Payments → Profile
Driver	Trips → Duty Slip / Trip Detail (OTP + KM capture) → Profile

7. Open Assumptions (flag if wrong)

- Personal Customers are assumed to always have app access in this phase (BRD mentions an "offline system for non-tech-savvy customers" as a separate later requirement, not in scope here).
- Corporate/Event login-splitting is assumed to be enforced at the account/token level (each of the 3 logins has its own credentials, same parent Customer ID).
- Vendor's 4 logins are assumed to work the same way — separate credentials, one Vendor ID.
- COD/cash-collection-by-driver is not explicit in the BRD Payment section; included as a light touch on the Driver payment-confirmation row — confirm if this is real or should be removed.